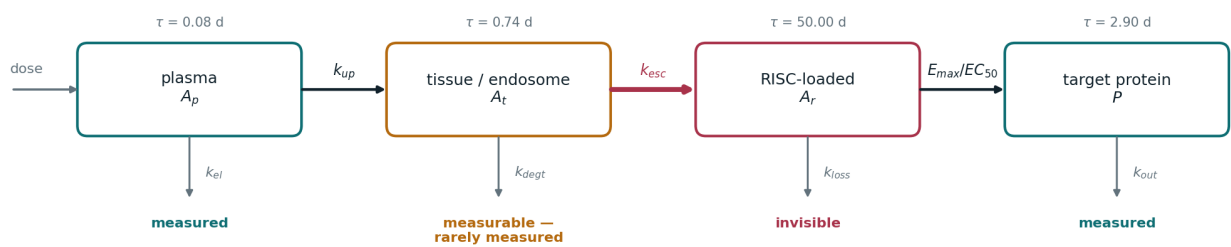


Identifiability of endosomal escape in a minimal siRNA model

What can be estimated, what cannot, and which experiment would change that — every number below is computed by `run_all.py`

1 · The model

Four compartments, following an siRNA from injection to effect. Plasma clears in hours; the RISC-loaded pool persists for months. That mismatch is the defining feature of the modality.



k_{esc} never appears alone: only as $(k_{esc} + k_{degt})$ in a pole, and inside $D k_{up} k_{esc} / EC_{50}$ in the gain

$dA_p/dt = -(k_{el} + k_{up}) \cdot A_p$	plasma	MEASURABLE
$dA_t/dt = k_{up} \cdot A_p - k_{esc} \cdot A_t - k_{degt} \cdot A_t$	tissue	MEASURABLE, usually not measured
$dA_r/dt = k_{esc} \cdot A_t - k_{loss} \cdot A_r$	RISC	INVISIBLE
$dP/dt = k_{syn} \cdot (1 - E_{max} \cdot A_r / (EC_{50} + A_r)) - k_{out} \cdot P$	protein	MEASURABLE

Parameter	Value	Meaning
k_{el}, k_{up}	2.50, 10.00 /d	Plasma elimination and tissue uptake; 80 % of dose reaches tissue
k_{esc}	0.135 /d	Endosomal escape — the productive route (10 % of tissue siRNA)
k_{degt}	1.216 /d	Nuclease degradation; competes with escape, and wins
k_{loss}	0.020 /d	RISC pool decay — 50 d time constant
E_{max}, EC_{50}	0.90, 0.0801	Efficacy and potency
k_{syn}, k_{out}	0.345, 0.345 /d	Protein synthesis and turnover; baseline = 1

STRUCTURE

The bracket in line 4 is an **E_{max} model**, not Michaelis–Menten. Same algebraic form, different meaning: concentration in, fractional *effect* out, multiplying a synthesis rate. That makes it an indirect-response structure.

Parameters are illustrative and fitted to nothing. EC_{50} is calibrated so the reference simulation reaches a 36.7 % nadir; everything else is set by hand.

2 · The identifiability problem

You observe the ends of the chain and not the middle. Protein is measurable and is what you care about. Plasma is measurable but gives only the *sum* $k_{el} + k_{up}$, never k_{up} on its own — and k_{up} sits in the gain.

The two compartments between them, where escape efficiency and potency live, are not routinely observed.

Consequence: k_{esc} , E_{max} and EC_{50} appear only through their joint effect on protein. A small amount of siRNA that suppresses strongly, and a large amount that suppresses weakly, produce **the same protein curve**.

THE POINT

Escape and potency are independent properties. Escape is a delivery property — conjugate, endosomal route, formulation. Potency is a pharmacology property — sequence, thermodynamics, chemical modification. Nothing links them, which is exactly why the data cannot separate them.

3 · What the algebra says

Laplace-transforming the linear-regime cascade and working in fractional suppression (which removes k_{syn}) gives the protein response to a bolus dose D:

$$dP(s)/P_0 = \frac{-D * k_{\text{out}} * k_{\text{up}} * (E_{\text{max}} * k_{\text{esc}} / EC_{50})}{(s + k_{\text{loss}})(s + k_{\text{out}})(s + k_{\text{degt}} + k_{\text{esc}})(s + k_{\text{el}} + k_{\text{up}})}$$

What a protein time course recovers from this is a **set of four time constants and one number**. k_{esc} appears twice — in the gain, where it is multiplied by E_{max}/EC_{50} and by the uptake rate, and in a pole, where it is added to k_{degt} . Neither occurrence isolates it.

Process	Time constant	Amplitude share	Still alive at day 28?
Plasma ($k_{\text{el}} + k_{\text{up}}$)	0.08 d	0.1 %	no
Tissue ($k_{\text{degt}} + k_{\text{esc}}$)	0.74 d	13.3 %	no
Protein turnover (k_{out})	2.90 d	49.9 %	no (0.01 %)
RISC decay (k_{loss})	50.00 d	36.7 %	yes

THE DISTINCTION

Amplitude share is a shape metric, not an information metric. The tissue pole — the only place k_{esc} appears on its own — carries 13.3 % of the amplitude, but on a monthly schedule its term has decayed to $4e-17$ of its initial value before the first sample is drawn. Sample at day 0.25 instead and 71 % of it is still there. **Practical identifiability is a property of the design, not of the model.**

4 · What is structurally identifiable

Practical identifiability is about noise and schedules. Before that comes a prior question: could k_{esc} be recovered from perfect, noise-free, continuously sampled data? It could not, and the reason is exact.

Rescale the upstream states by $u = A_p/D$, $v = A_t/(D \cdot k_{\text{up}})$ and $w = A_r/(D \cdot k_{\text{up}} \cdot k_{\text{esc}})$. What is left carries the three poles and nothing else — D, k_{up} and k_{esc} cancel:

$$\begin{aligned} u: & a \cdot u(t) + \text{Derivative}(u(t), t) \\ v: & b \cdot v(t) - u(t) + \text{Derivative}(v(t), t) \\ w: & k_{\text{loss}} \cdot w(t) - v(t) + \text{Derivative}(w(t), t) \end{aligned}$$

And the E_{max} term is homogeneous of degree zero in (A_r, EC_{50}) , so it sees those parameters only through $\psi = D \cdot k_{\text{up}} \cdot k_{\text{esc}} / EC_{50}$ (verified symbolically; residual 0). Nine parameters therefore reach the observables through seven functions:

$k_{el} + k_{up}$	$k_{esc} + k_{degt}$	k_{loss}	k_{out}	k_{syn}	E_{max}	ψ
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The level sets of that map are a two-parameter group. Take any $\alpha, \beta > 0$ and send k_{up} to $\alpha \cdot k_{up}$, k_{esc} to $\beta \cdot k_{esc}$, EC_{50} to $\alpha \cdot \beta \cdot EC_{50}$, with k_{el} and k_{degt} absorbing the difference. Plasma and protein do not move:

	As simulated	Transformed	
k_{esc}	0.1351 /d	0.5405 /d	4 × larger
Escape fraction	10 %	40 %	a different delivery story
EC_{50}	0.0801	0.3683	compensating
Protein curve	—	differs by 3e-11	integrator tolerance
Plasma curve	—	differs by 2e-13	integrator tolerance

THE STRUCTURAL RESULT

k_{esc} is not structurally identifiable from plasma and protein — not at any sampling density, not with any instrument. A four-fold difference in escape and a four-fold difference in escape fraction produce the same data to eleven decimal places. The practical identifiability analysis that follows is therefore not a story about insufficient data; it is a story about what an additional measurement buys once you accept that this one cannot be rescued by measuring harder.

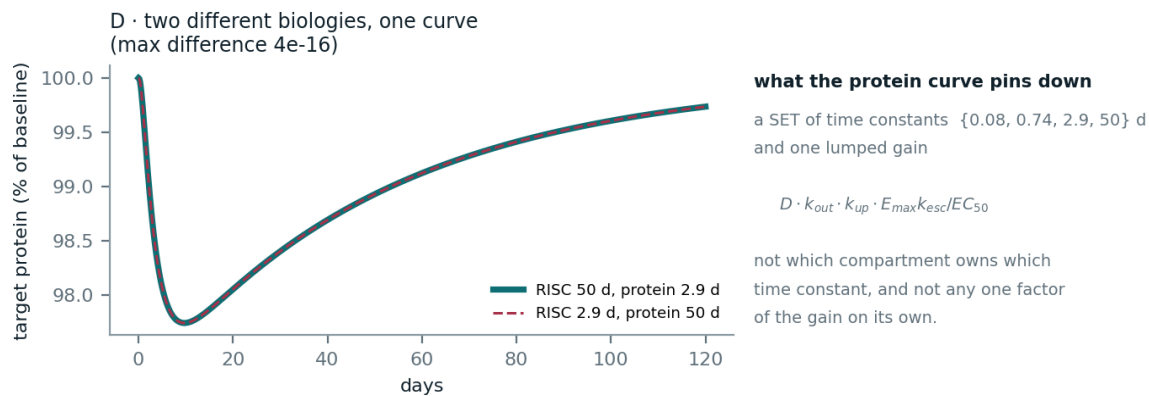
Confirming it numerically: the rank of the relative sensitivity map, which counts how many independent directions in parameter space the data can see. Rank is taken at the largest gap in the singular value spectrum rather than at a fixed cutoff, because the smallest singular value moves by two orders of magnitude with the finite-difference step while the gap does not.

Design	Free parameters	Rank	Unresolved directions
plasma + protein	9	7	2
plasma + protein + total tissue siRNA	9	8	1
the above, with k_{degt} fixed by a stability assay	8	8	0

The two unresolved directions are the group generators, recovered to three decimals from the singular value decomposition: $k_{el} -0.95$, $k_{up} +0.24$, $k_{esc} -0.08$, $EC50 +0.16$; $k_{esc} +0.71$, $k_{degt} -0.08$, $EC50 +0.70$. A check that depends on no threshold at all: applying the analytic generators to the sensitivity matrix gives residuals of 4e-09 and 1e-09 relative — integrator noise — while the first of the two rises to 4e-02 once tissue siRNA is observed, which is precisely the direction that measurement resolves.

4b · And in the linear regime, which pole is which

k_{loss} and k_{out} survive the analysis above as separately identifiable, but only because of the saturation nonlinearity. Drop to the linear regime — where ϕ is defined — and the transfer-function denominator becomes symmetric under permutation of its roots while the numerator is a single number, so any reassignment of time constants to compartments can be absorbed by rescaling the gain. A 50-day RISC pool feeding a 2.9-day protein, and a 2.9-day RISC pool feeding a 50-day protein, give curves identical to 4e-16 relative.



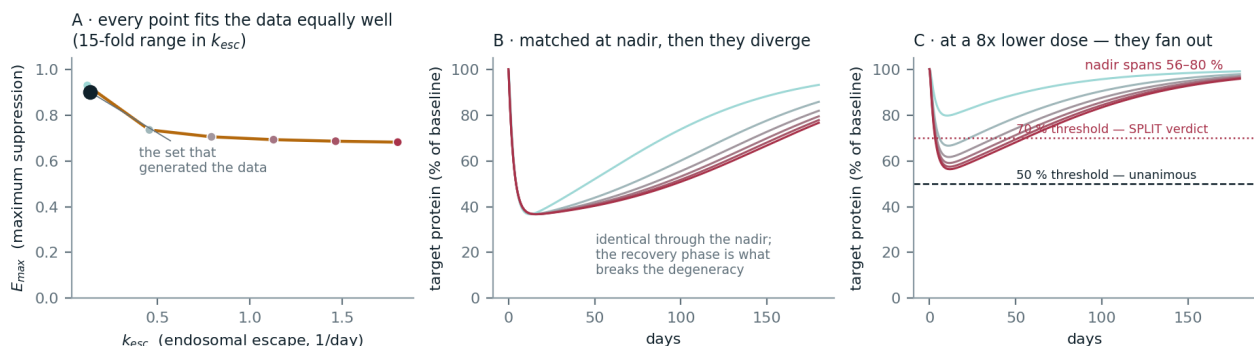
Left: the two assignments, plotted on top of each other, inside the linear range. Right: what the protein curve determines there.

WHAT THIS COSTS

So durability is inferable only from the part of the response that the linear approximation throws away. At the reference dose a refitted swapped model is 14 percentage points of baseline away, so the curvature does carry it — but an analysis conducted entirely in the linear regime cannot tell a persistent RISC pool from a slow-turnover target. Independent measurement of protein turnover settles it either way.

5 · Does the degeneracy matter? Walk the set

Not knowing a parameter is only a problem if the decision depends on it. The test: fix the fit, walk the parameter combinations that reproduce it, re-run the untested prediction on every point, and look at the spread of *verdicts*.



Left: every point reproduces the observed 36.7 % nadir — a 15-fold range in k_{esc} . Centre: the curves are identical through the nadir, then diverge during recovery. Right: at a 8-fold lower dose the nadir spans 56–80 % and they fan out.

Decision rule (at the lower dose)	Verdicts across the set	Conclusion
Is nadir ≤ 50 % of baseline?	FAIL $\times 6$	Unanimous — the degeneracy is irrelevant to this question
Is nadir ≤ 70 % of baseline?	PASS $\times 5$, FAIL $\times 1$	Split — the decision rests on what you cannot resolve

THE METHOD

Same ignorance, different consequence. Most modelling anxiety about unknown parameters dissolves once you check whether they actually move the answer — and the ones that survive that check are the ones worth an experiment.

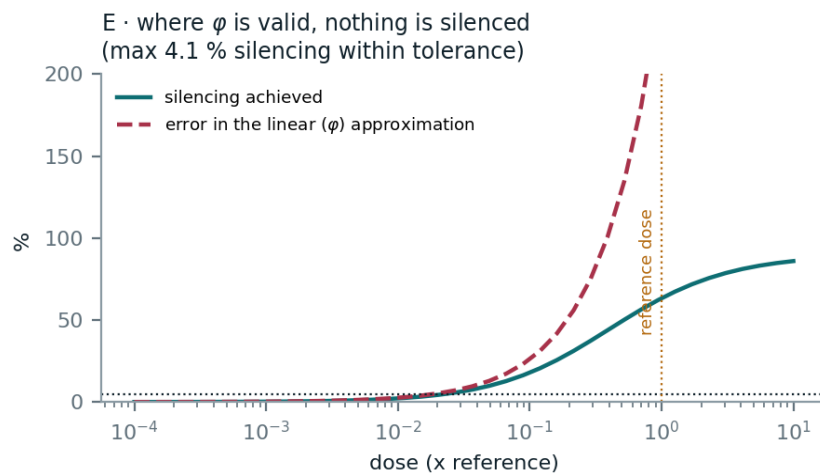
A CAVEAT ON THE METHOD

The set walked above is an **iso-nadir set**: one scalar constraint, which is what you have if the nadir is all you measure. It is not the likelihood-flat manifold of the full model, which additionally holds the tissue pole fixed. The two are different objects and the distinction matters: the curves separate during recovery precisely because the iso-nadir set does not preserve the tissue pole. **Extending the sampling window is itself an answer.**

6 · What is measurable: phi, and where it holds

$$\phi = E_{\max} \cdot k_{\text{esc}} / EC_{50} = 1.52 \quad \text{total silencing produced per unit dose}$$

ϕ is defined in the linear approximation, where $E_{\max} A_r / (EC_{50} + A_r)$ is approximately $(E_{\max} / EC_{50}) A_r$. That approximation needs $A_r \ll EC_{50}$. Deep silencing needs the opposite. Those two requirements are in direct conflict, and the conflict is quantitative:



At the reference dose, peak A_r is $2.8 \times EC_{50}$ — 74 % of the way to the ceiling — and the linear approximation misstates the depth of silencing by 257 %.

THE CATCH

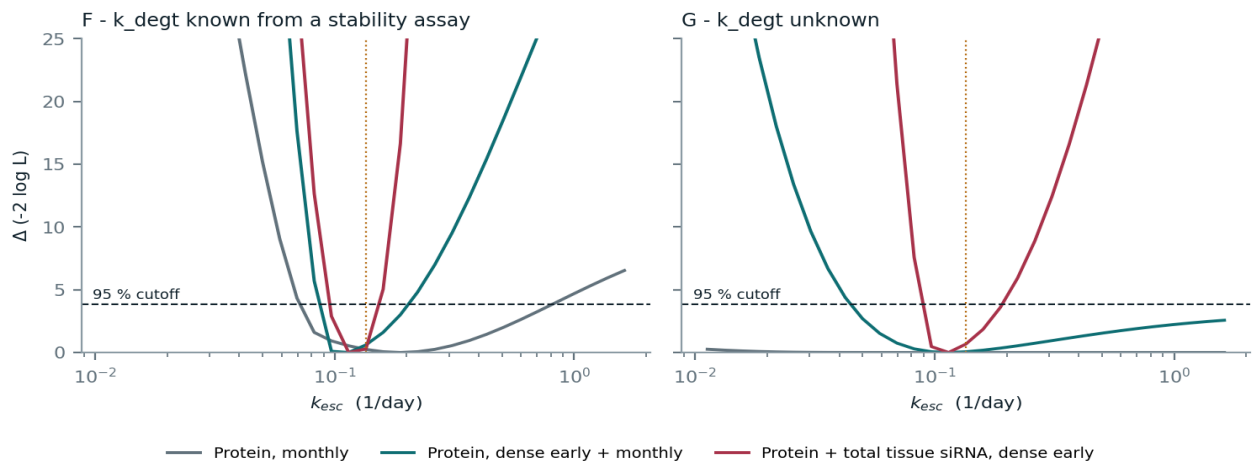
The validity window may sit below the noise floor. Holding the linear approximation to 5 % relative error caps the achievable silencing at 4.1 % of baseline — below a typical 10 % assay CV. In this parameterisation ϕ is a well-defined quantity that is only valid at doses where there is nothing to measure. That is a structural feature of an $E_{\max}$ model, not an artefact of these numbers: any parameter set that silences deeply is operating near the ceiling.

phi answers	phi cannot answer
Which of six candidates gives most silencing per milligram, at matched dose and tissue	Whether an underperformer has a delivery or a sequence problem
How dose–response scales, inside the validity window	Whether a conjugate working in liver will work in muscle
Whether a chemistry change improved overall efficiency	What EC_{50} is — it is confounded with $k_{\text{up}} \cdot k_{\text{esc}}$, though $E_{\max}$ itself is recoverable from the saturation curvature

7 · Breaking the degeneracy, and what each experiment buys

Each confounded parameter can be isolated in a system where you control what the in vivo model cannot observe. The question is which one to pay for. Profile likelihood on simulated data answers it: for each

sampling design, profile k_{esc} and read off the 95 % interval.



A flat profile is practical non-identifiability. Left: k_{deg} known from a stability assay. Right: k_{deg} unknown — the honest starting point.

Sampling design	k_{deg} known	k_{deg} unknown
Protein, monthly	11.3-fold	over 144-fold
Protein, dense early + monthly	2.3-fold	over 36-fold
Protein + total tissue siRNA, dense early	1.6-fold	2.1-fold

WHAT THE NUMBERS SAY

The experiment that matters is the tissue measurement, not the sampling schedule. With k_{deg} unknown, moving protein sampling earlier narrows k_{esc} from more than 144-fold to more than 36-fold — still useless. Adding total tissue siRNA on the same early schedule brings it to 2.1-fold. Dense early protein sampling is not a substitute for measuring the compartment the pole belongs to.

Parameter	Experiment	Why it works
E_{max}, EC_{50}	Transfection across a wide concentration range; measure target mRNA and protein	Transfection bypasses escape entirely. Deliver straight to the cytosol and k_{esc} leaves the chain, so what you measure is pharmacology alone
k_{deg}	Nuclease stability assay in vitro	Fixes the other term in the tissue pole
k_{out}	Independent protein turnover measurement	Resolves the pole-assignment ambiguity in section 4
k_{esc}	Total tissue siRNA on a dense early schedule; subcellular fractionation or Ago2 RNA immunoprecipitation for the escaped fraction	Observes the compartment where the tissue pole lives, rather than inferring it from downstream protein

CAVEAT

Two honest limits. In vitro EC_{50} does not transfer cleanly to in vivo concentrations without an intracellular reference. And transfection is not GalNAc-mediated uptake, so the escape route differs from the real one — which is precisely what microphysiological systems are trying to close.

8 · Why k_{esc} is worth the effort

If the problem is	Then	Send it back to
Low k_{esc}	Enough drug arrives; little escapes the endosome	Delivery — conjugate, formulation, chemistry
Low E_{max}/EC_{50}	Plenty escapes; the silencing is weak	Sequence — design and modification pattern

Without k_{esc} you cannot tell them apart, so you are guessing which team to send an underperforming compound to. It also gates extrapolation: GalNAc works in liver because ASGPR is hepatocyte-specific, and whether a new conjugate reaches muscle or CNS depends on whether enough escapes there.

SUMMARY

The one-line version. The reason k_{esc} is worth measuring is not precision — it is that it tells you whether an underperforming compound is a delivery problem or a sequence problem. Those go back to different teams.

Parameters are illustrative and no proprietary or clinical data was used. The transferable content is the structure of the argument, not the numbers. Every figure and every value in the tables above was produced by `run_all.py` (84 s) and read from `results/results.json`, generated 2026-09-16 01:13:10 UTC.